

Analyzing Secondhand Car Sales Data with Supervised and Unsupervised Learning Models

Abstract

This study investigates the application of supervised and unsupervised machine learning models for the analysis and forecasting of used automobile prices in the United Kingdom. A synthetic dataset containing 50,000 entries which includes both numerical and categorical features such as mileage, engine size, fuel type, and year of manufacture was used. Various regression models were created including simple Linear Regression, Multiple Linear Regression, Random Forest and Artificial Neural Network (ANN). The ANN slightly outperformed Random Forest by a small margin and achieved the best predictive performance ($R^2 = 0.9992$). To identify patterns in the dataset, clustering analysis was also done using DBSCAN and k-Means. DBSCAN only performed well in one feature combination, while k-Means produced more stable and comprehensible clusters. Overall, the study highlights how important feature integration, model selection, and metric-based evaluation are to forecasting car prices.

1. Introduction

Accurate pricing models benefit both buyers and sellers in the used automobile market, which is a significant segment of the automotive industry. Machine learning provides strong methods for analyzing large datasets and making accurate price predictions. Both linear and nonlinear regression models are useful for predicting prices using features like mileage and production year, according to earlier research on the Croatian used car market (Bukvić et al., 2022). Combining categorical and numerical variables to increase model accuracy is still challenging, though. Furthermore, clustering techniques can be used to uncover meaningful vehicle segments, supporting strategic inventory and marketing decisions, as shown by market segmentation studies such as Farhan et al. (2024). Using a synthetic UK used car dataset, this study predicts prices and identifies trends using both supervised and unsupervised models.

2. Methodology

2.1 Dataset

The analysis used a fictitious dataset of 50,000 UK used automobile sales that includes both categorical features (manufacturer, model, fuel type) and numerical features (year of manufacture, mileage, engine size, and price). The dataset was checked to know if there were missing values, but none was found present. This enabled straightforward modeling.

2.3 Exploratory Data Analysis

To understand feature distributions, identify outliers, and evaluate correlations between variables, exploratory data analysis or EDA, was carried out. Plots were created to explore trends across variables and, a seaborn heatmap was used to calculate and display a correlation matrix. This made it possible to determine which factors had the greatest influence on car price prediction.

Mileage and year of manufacture had the strongest correlation ($r = -0.81$), according to the correlation heatmap (Figure 2.1), suggesting that newer cars typically have lower mileage. In addition, price demonstrated a moderately negative correlation with mileage ($r = -0.63$) and a strong positive correlation with year ($r = +0.71$), confirming that newer, less-driven cars typically fetch higher resale prices. Price and engine size had a weaker correlation ($r = +0.40$), indicating a more constrained impact.

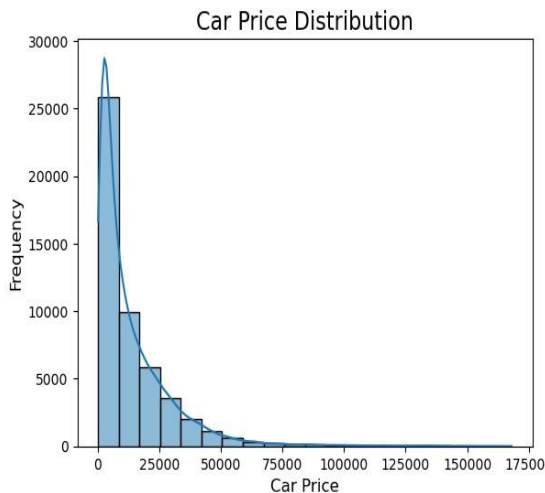


Fig 2.1 Histogram plot showing price distribution

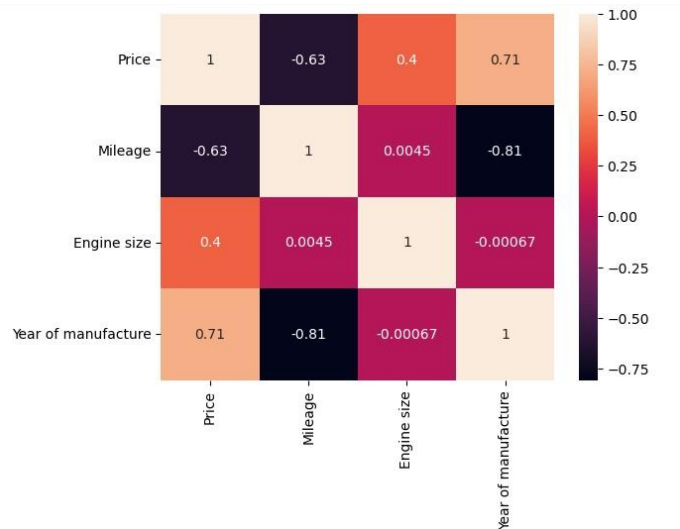


Fig 2.2 Heatmap showing relationships between numerical features

2.2 Data Preprocessing

- Label encoding / one hot encoding was applied to categorical variables such as manufacturer, model, and fuel type. This transformed them into dummy variables for both supervised and unsupervised modeling.
- Standardization (zero mean, unit variance) was applied to numerical variables, particularly before clustering and neural network training to guarantee scale consistency

2.3 Supervised Learning Models

Three classes of supervised models were trained to predict the car price:

- **Single-Feature Regression Models:** Linear and polynomial regression were trained independently for each numerical feature (year, mileage, engine size) to predict car prices.
- **Multiple Linear Regression:** All numerical variables were combined as input features and trained multiple linear regression model to determine whether multiple inputs improved predictive accuracy.
- **Random Forest Regressor:** An ensemble model was trained using all numerical features as input to determine whether it improved accuracy when compared to multiple linear regression model. Later extended to include categorical variables. This approach is common in used car price prediction research (Sampath et al., 2024).
- **Artificial Neural Network (ANN):** An ANN was trained with scaled numerical features and one-hot encoded categorical features as input. Two hidden layers (64, 64 neurons) with ReLU activation and one linear output neuron for regression made up the default architecture. The model was built using the Adam optimizer and trained over 200 epochs with a learning rate of 0.001, a patience of 20, and dropout rate of 0.2. After increasing the first layer to 128 units and adding a third layer, performance was enhanced by adjusting hyperparameters. When compared to other models (like Random Forest and Linear Regression Models), the ANN produced competitive results and performed better after hyperparameter tuning.

Standard regression measures that are frequently used in price prediction tasks, such as R2 Score, Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE), were employed to assess the supervised models. Lin (2025) demonstrated in their study on used car price prediction that this evaluation method is in line with best practices in machine learning-based valuation modeling.

2.4 Unsupervised Learning Models

To investigate latent groupings in the data, two clustering algorithms were employed:

- **k-Means Clustering:**
 - Applied to six numerical variable combinations: mileage, year of manufacture, engine size, and price.
 - The optimal number of clusters, k, was determined using the Elbow method, which is based on the within-cluster sum of squares. The Davies-Bouldin Index and Silhouette Coefficient were then used to confirm the result.
- **Density-Based Spatial Clustering or DBSCAN:**
 - Applied using fixed parameters of $\text{eps} = 0.2$ and $\text{min_samples} = 3$ across all six combinations.
 - Chosen to explore whether density-based clustering produces more stable or significant clusters with variable input sets. ○ Evaluated with same metrics as k-Means: Silhouette and Davies-Bouldin scores.

2.5 Model Comparison

- **Supervised models** were compared by choosing the best-performing configurations: (a) best performed single-feature polynomial regression, (b) multiple linear regression, (c) Random Forest (with both numerical and categorical variables), and (d) ANN. Performance metrics (MAE, MSE, R² Score, RMSE) were tabulated and discussed.
- **Clustering models** (k-Means vs. DBSCAN) were compared based on the best combination of objective metrics: Silhouette, Davies-Bouldin and consistency across input features.

3. Results

3.1 Regression Models Based on Single Numerical Feature

To determine which individual numerical feature best predicts car price, separate regression models were trained using the numerical features contained in the dataset which includes Mileage, Engine Size and Year of Manufacture.

Each of the features was used in both linear and polynomial regression models. The model's performance was evaluated using the R2 score, Mean Absolute Error (MAE), Mean Squared Error (MSE) and Root Mean Squared Error (RMSE). Table 3.1 shows the summary of the results:

Table 3.1: Single Feature Metrics and Performance Summary

Input Feature	Model Type	MAE	MSE	RMSE	R2 Score
Mileage	Simple Linear	7964.78	162468566.87	12746.32	0.4013
	Polynomial	6409.91	129620312.16	11385.09	0.5224
Engine size	Simple Linear	10817.49	230499154.45	15182.20	0.1506
	Polynomial	10807.26	230326166.00	15176.50	0.1513
Year of Manufacture	Simple Linear	7031.04	132678999.95	11518.64	0.5111
	Polynomial	5387.11	105993894.20	10295.33	0.6094

Year of manufacture, when modeled using polynomial regression, performed better than the other single-feature models by achieving the highest R^2 score and lowest error values which indicates that vehicle age is the most accurate predictor among other numerical features.

Polynomial regression continuously outperformed linear models across all numerical input features, revealing non-linear relationships between each variable and car price. The actual price values were closely matched with the polynomial fit for the year of manufacture. These results align with earlier research demonstrating that a vehicle's age significantly influences its resale value (Gao, 2024; Totakura et al., 2021).

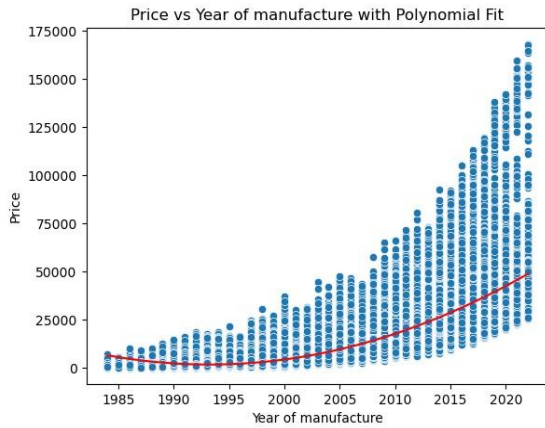


Fig 3.1: Polynomial Regression (Year of Manufacture)

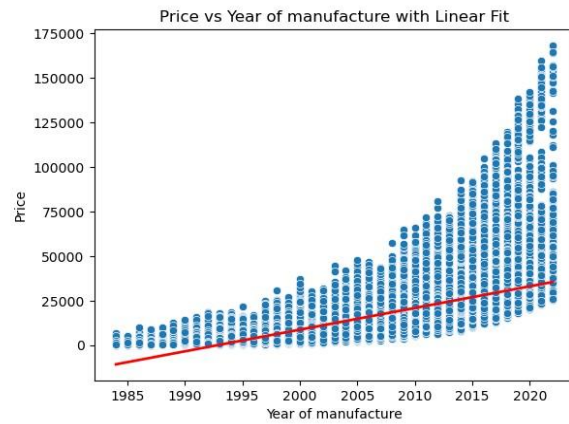


Fig 3.2: Simple Linear Regression (Year of Manufacture)

3.2 Linear Regression Model Based on Multiple Numerical Features

To evaluate whether the including multiple numerical features improves price prediction, two models were trained using mileage, engine size, and year of manufacture as input variables:

- Multiple Linear Regression
- Random Forest Regressor

These were compared against the best-performing single-feature model from section 3.1 (Polynomial Regression using year of manufacture). The table below summarizes the performance:

Table 3.2: Multiple Features Evaluation Metrics

Model	Input Features	MAE	MSE	RMSE	R ² Score
Polynomial Single Feature (Year)	Year of Manufacture	5387.11	105993894.20	10295.33	0.6094
Multiple Linear Regression	Mileage, Engine Size, Year	6091.46	89158615.76	9442.38	0.6715
Random Forest Regressor	Mileage, Engine Size, Year	2294.47	20159131.52	4489.89	0.9257

Both multiple-feature models outperformed the single-feature model, with the Random Forest Regressor obtaining the highest predictive accuracy. This findings highlights that:

- Car price is affected by a combination of numerical features, not just one.
- Ensemble models, like Random Forest, perform better in structured data problems because they are better at capturing non-linear interactions and feature combinations. (Totakura et al., 2021; Tin et al., 2024)

3.2.1 Performance Analysis

The actual vs. predicted price plot (see Figure 3.3) demonstrated a strong diagonal alignment, suggesting that the model did well with respect to prediction, particularly for low and mid-range vehicles. However, as data points became more dispersed, predictions for more expensive cars became less accurate. This implies that rare or expensive cars are difficult to generalize, most likely because of the dataset's smaller sample size (Totakura et al., 2021).

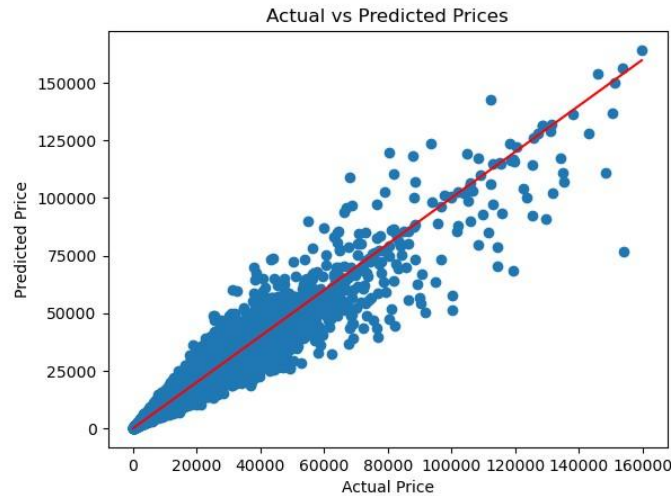


Fig 3.3: Actual vs. Predicted Car Prices using Random Forest Regressor for numerical features

3.3 Regression Model Based on All Relevant Features

To determine whether adding categorical variables improves car price prediction, a Random Forest Regressor was trained using both the numerical (mileage, engine size, year of manufacture) and categorical (fuel type, manufacturer, model) features. One-hot encoding was used as a preprocessing method to transform categorical variables into numerical format. Then the model's performance was compared with that of the Random Forest that was trained solely using numerical features (from section 3.2).

Table 3.3: Random Forest Regressor Evaluation Metrics

Model	Input Features	MAE	MSE	RMSE	R ² Score
Random Forest (numerical only)	Mileage, Engine Size, Year	2294.47	20159131.52	4489.89	0.9257
Random Forest (numerical + categorical)	Mileage, Engine Size, Year + Fuel Type, Manufacturer, Model	292.19	413269.54	642.86	0.9985

The results demonstrate that prediction accuracy was significantly increased by the addition of categorical variables. The R2 score increased from 0.9257 to 0.9985, while MAE and RMSE dropped by more than 50%. The impact of brand-related factors on resale value is reflected in this improvement.

3.3.1 Performance Analysis

A plot comparing actual vs. predicted car prices was created for the Random Forest Regressor trained on all numerical and categorical features (see Figure 3.4). High model accuracy was indicated by the majority of data points strongly aligning along the diagonal line. In every price range, including more expensive cars, the expected and actual prices were very similar. The model's ability to identify complex relationships in the dataset and generalize effectively is validated by its consistency across low, mid, and high price ranges.

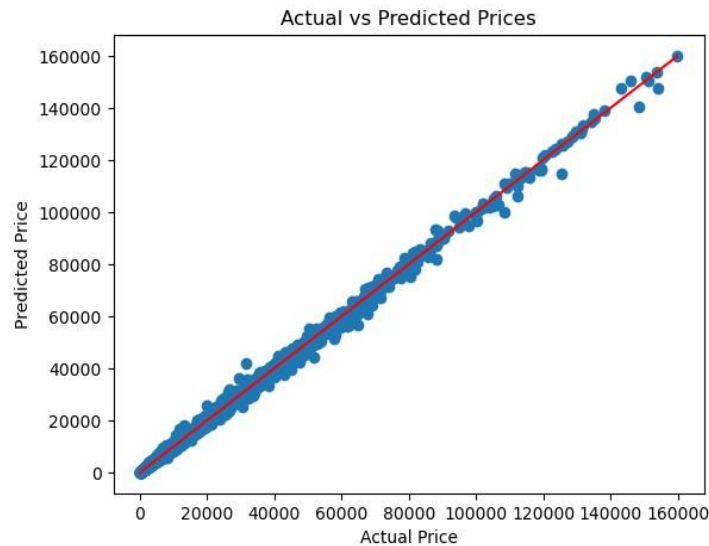


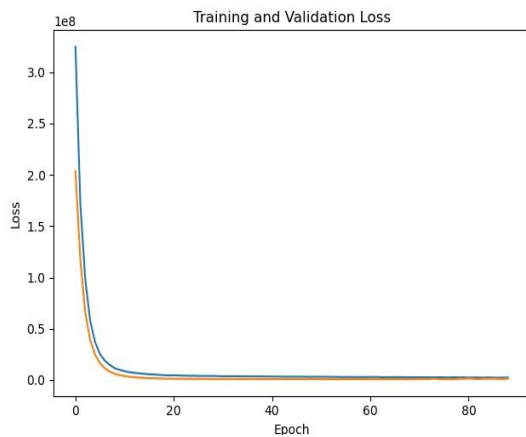
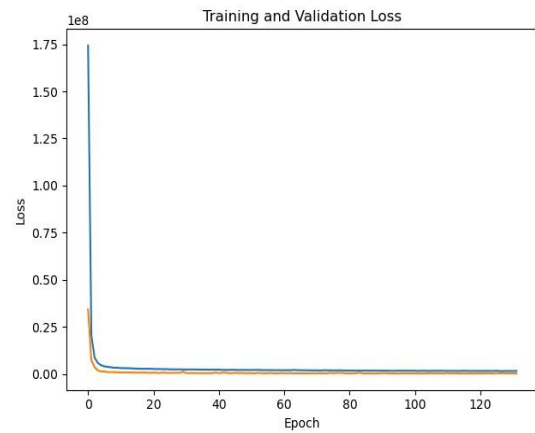
Fig 3.4: Actual vs. Predicted Car Prices using Random Forest Regressor for numerical and categorical features

3.4 Artificial Neural Network (ANN) Model

The number of hidden layers and other parameters were changed to optimize the model architecture and training parameters. The final tuned version of the ANN outperformed when considering the evaluation metrics: R2, Mean Absolute Error (MAE), Mean Squared Error (MSE) and Root Mean Squared Error (RMSE), as indicated in Table 3.4, even though some configurations performed worse than some supervised models.

Table 3.4: ANN Evaluation Metrics (Hyperparameter Tuning)

Hidden Layer	Layer Units	Learning Rate	Dropout Rate	MAE	MSE	RMSE	R2 Score
2	64, 64	0.001	0.2	817.84	1345688.75	1160.04	0.9950
3	64, 64, 64	0.001	0.2	1165.69	2260863.50	1503.62	0.9917
	64, 64, 64	0.01	0.2	2188.93	9470066.00	3077.35	0.9651
	64, 64, 64	0.0001	0.2	403.91	519793.56	720.97	0.9981
	64, 64, 64	0.0001	0.1	281.36	277329.91	526.62	0.9990
	128, 64, 64	0.001	0.3	315.44	223867.19	473.15	0.9992

*Fig 3.5: ANN, Two hidden layers**Fig 3.6: ANN, Three hidden layers (Overall best)*

3.5 Best Model for Price Prediction

Based on the analysis findings, the ANN model was found to be the most effective for predicting second-hand car prices. Among all supervised models tested, the final tuned ANN had the highest accuracy for prediction, with R^2 score of 0.9992, MAE of 315.44, and RMSE of 473.15 as seen in Table 3.5.

Table 3.5: Best Models Performance Comparison

Model	MAE	MSE	RMSE	R ² Score
Simple Linear Regression	7031.04	132678999.95	11518.64	0.5111
Polynomial Regression	5387.11	105993894.20	10295.33	0.6094
Multiple Linear Regression	6091.46	89158615.76	9442.38	0.6715
Random Forest Regressor	292.19	413269.54	642.86	0.9985
Artificial Neural Network	315.44	223867.19	473.15	0.9992

The ANN's superior performance can be attributed to its capacity to represent intricate, nonlinear relationships and interactions between a large number of categorical and numerical features. This outcome is in line with previous studies by Khotimah et al. (2025) and Bergmann et al. (2025), who also discovered that ANN models outperform Random Forest and linear models when it comes to predicting used car prices from high-dimensional data.

Therefore, the ANN model is recommended as the best supervised learning approach for this problem.

3.6 Clustering Analysis Using k-Means

k-Means clustering was applied to six combinations of the numerical features in the dataset. The performance of each configuration was assessed using the Silhouette Coefficient and DaviesBouldin Index. The summary of the results shown in Table 3.6 below.

Table 3.6: K Means Clustering Evaluation Metrics

Feature Combination	Optimal k	Silhouette Score	Davies-Bouldin Index
Price + Mileage	3	0.478	0.688
Price + Year	3	0.514	0.687
Price + Engine Size	5	0.432	0.755
Milage + Year	2	0.533	0.659
Milage + Engine Size	3	0.450	0.777
Year + Engine Size	3	0.460	0.753

Among all feature combinations tested, the Mileage and Year of Manufacture combination gave the best clustering results, with a Silhouette Score of 0.533 and a Davies-Bouldin Index of 0.659 using k = 2 clusters.

The optimal number of clusters (k) for each configuration was determined using the elbow method and confirmed using both evaluation metrics. Most combinations showed stable results around $k = 3$, but the Mileage and Year combination showed better cohesion and separation with $k = 2$, indicating the presence of two distinct vehicle categories.

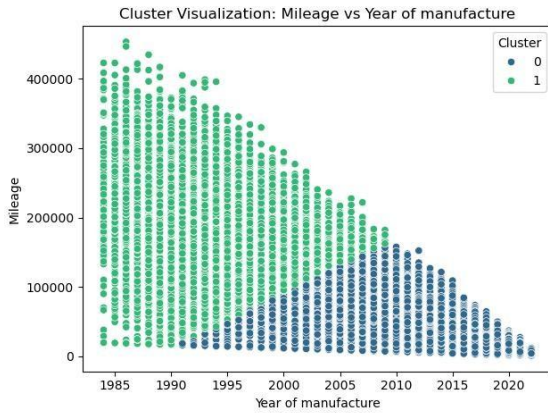


Fig 3.7: K Means Clustering, Mileage and Year

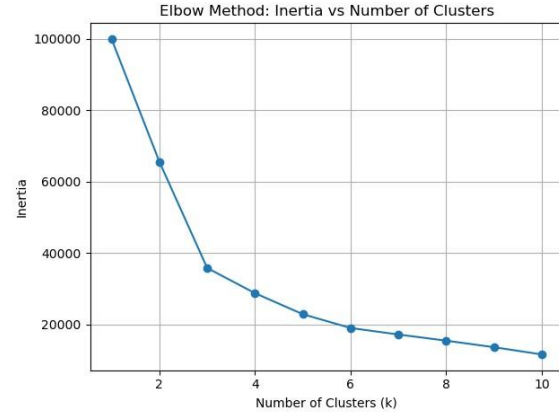


Fig 3.8: Elbow plot, Engine size and Mileage

3.7 DBSCAN Clustering

The DBSCAN (Density-Based Spatial Clustering of Applications with Noise) algorithm was also applied to the same six feature combinations with fixed parameters: $\text{eps} = 0.2$ and $\text{min_samples} = 3$ across combinations in order to assess whether alternative clustering methods could perform better than k-Means. The Davies-Bouldin Index and the Silhouette Coefficient were used to evaluate the performance.

Table 3.7: DBSCAN Clustering Evaluation Metrics

Feature Combination	Silhouette Score	Davies-Bouldin Index
Price + Mileage	0.8178	0.1570
Price + Year	0.6763	1.4952
Price + Engine Size	0.0608	2.6414
Milage + Year	0.6316	0.2529
Milage + Engine Size	0.0467	3.6998
Year + Engine Size	0.0581	5.0472

The combination of Price and Mileage gave the best result for DBSCAN, with a Silhouette Score of 0.8178 and a Davies-Bouldin Index of 0.1570 which is better than its performance on other feature pairs.

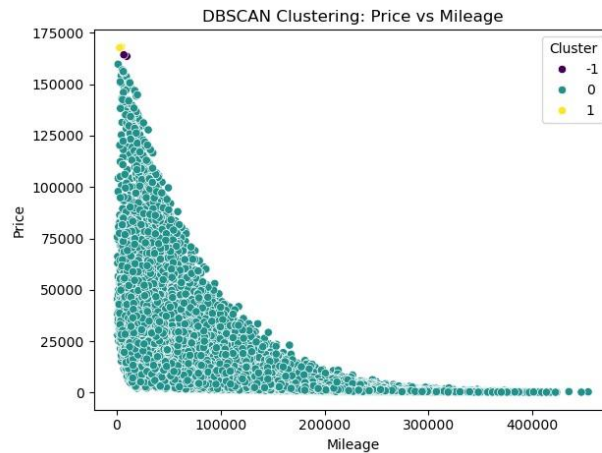


Fig 3.8: DBSCAN Clustering, Price and Mileage

Although DBSCAN outperformed k-Means in this setup, its general applicability in this dataset is limited by its uneven performance across feature combinations, as well as its sensitivity to ϵ and cluster density. However, for structured data with roughly spherical cluster shapes, k Means produced reliable clusters for the majority of input combinations (Mikrou et al., 2025).

4. Discussion

4.1 Supervised Learning Insights

The best predictor of car price was the Artificial Neural Network (ANN) model, which performed better than the Random Forest, Multiple Linear Regression, and Polynomial Regression models. In high-dimensional data fields, the final tuned ANN showed an exceptional capacity to capture complicated, non-linear interactions, achieving R^2 of 0.9992, MAE of 315.44, and RMSE of 473.15. This is consistent with research by Lin (2025), who found that in tasks involving the forecasting of used car prices, Random Forest and ANN models typically perform better than simple regression techniques.

The ANN slightly outperformed the Random Forest Regressor, despite the latter's strong performance ($R^2 = 0.9985$). Practically speaking, when the highest level of precision is needed, especially at mid-to-high price points, the ANN's slight improvement is justified. When interpretability and training efficacy are important considerations, Random Forest is still a fantastic alternative.

4.2 Feature Impact and Interpretability

The year of manufacture, when modeled independently, proved to be the most successful predictor, suggesting that the age of the vehicle is a significant factor in price. Year, mileage, and

engine size combined together, however, greatly increased predictive accuracy, suggesting that multi-feature models are better at capturing price variance.

4.3 Clustering Analysis Insights

k-Means generated continuous and interpretable clusters, especially when Mileage and Year were combined (Silhouette = 0.533, DBI = 0.659), differentiating between newer low-mileage from older high-mileage cars.

With fixed parameters, DBSCAN was unstable for all combinations but performed best with Price and Mileage (Silhouette = 0.8178, DBI = 0.157). This demonstrates its well-known parameter sensitivity and propensity to incorrectly label points in structured data (Kossakov et al., 2024).

5. Conclusion

This study showed how well machine learning models can detect trends in vehicle attributes and forecast used car prices. The Random Forest model performed well and was easier to understand, but the Artificial Neural Network model had the highest predictive accuracy. Across all feature combinations, k-Means outperformed DBSCAN in clustering.

Overall, the findings emphasize how crucial it is to apply both exploratory and predictive models, as well as to combine numerical and categorical features, in order to understand complex datasets.

References

Bukvić, L., Pašagić Škrinjar, J., Fratrović, T. and Abramović, B., 2022. Price prediction and classification of used-vehicles using supervised machine learning. *Sustainability*, 14(24), p.17034.

Farhan, M. and Heikal, J., 2024. Used Car Customer Segmentation Using K-Means Clustering Model With SPSS Program: Case Study Caroline. Id. *Jurnal Indonesia Sosial Sains*, 5(03), pp.543-559.

Sampath, O., gari Anusha, A. and Student, M.C.A., Prediction of Used Car Prices Using Artificial Neural Networks and Machine Learning.

Totakura, S.S.G.S.N. and Kosuru, H., 2021. Comparison of Supervised Learning Models for predicting prices of Used Cars.

Gao, J., 2024. Second-hand car price prediction based on multiple linear regression and random forest. *Theoretical and Natural Science*, 52, pp.31–40.

Tin, T.T., Kwoh Lik Xun, J., Bruse, G., Pong Soon Hui, J., Chee, L.W., Hun, C.K., Aitizaz, A., Khattak, U.F. and Salau, A.O., 2024. Car Depreciation Price Prediction Using Multiple Machine Learning Algorithms. *Pakistan Journal of Life & Social Sciences*, 22(2).

Khotimah, E.K., Swasono, D.I. and Fajarianto, G.W., 2025. Forecasting Used Car Prices Using Machine Learning. *IT Journal Research and Development*, 9(2), pp.123-139.

Bergmann, S. and Feuerriegel, S., 2025. Machine learning for predicting used car resale prices using granular vehicle equipment information. *Expert Systems with Applications*, 263, p.125640.

Mikrou, I. and Sapidis, N.S., 2025, April. A Systematic Evaluation of Clustering Algorithms Against Expert-Derived Clustering. In *Operations Research Forum* (Vol. 6, No. 2, p. 55). Cham: Springer International Publishing.